MIS Data Analytics Case Competition

Data Provided:

- City-collected traffic volumes from around Norman for five, non-consecutive years
- Collision data for ten years
- Network capacity information from ACOG
- GIS map of Norman
    - One layer is the Ward Maps
    - Aerial photography can be useful in determining the number of lanes
    - Although maps were provided, the best way to access the maps would be to go to Normanok.gov
        - Click on "For Residents & Visitors"
        - Click on "City Maps"
        - From the "City Maps" menu, you can click on the "Interactive Map" to see an aerial map of the city for use in determining the number of lanes on a segment
        - From the "City Maps" menu, you can click on "City Council Ward Map" to view the entire city broken into each of the eight wards or to view each ward independently.

Tasks

Traffic Volumes

- To begin, you need to divide the roadways in Norman up into segments for analysis. As guidance, we will suggest that each segment should be at least one mile in length to make sense as an improvement project.
- It is entirely possible that the same roadway may be in multiple spreadsheets by year. In the event this happens, you should use the highest traffic volume available. For example, if the volume in 2019 was 18,000 and the volume in 2025 was 20,000, you should plan to use 20,000 in your calculations.
- If a segment falls on a ward boundary, count the segment in both wards. Should the segment progress toward a recommended project, the city will determine how to treat it.
- From the aerial maps, determine the number of lanes associated with each traffic count
- Take the traffic counts (the volume) and determine a volume to capacity ratio (v/c) for each traffic count (some locations may be counted more than one time, so you will want to use the highest volume)

- Divide the segments into groups depending on which City ward they fall in
- Rank the v/c for the segments in each Ward (this will be 8 separate lists)

Collisions

- You should have the city roads divided into segments from your traffic volume portion of the exercise.
- Here is a major hint: any collisions that do not coincide with a segment where traffic volumes were collected, it should not proceed further in your decision-making.
- Map all the collisions to determine what segment they fall into
    - You could choose to start with the latitude and longitude provided for each collision. We think this is the latitude and longitude of the nearest intersection and may have reservations about the accuracy of these coordinates.
    - These points would need to be checked against the information relative to distance and direction from the referenced intersection to get the proper location for each collision. This would be the best approach to make sure you map the collision location in the proper segment.
- Analyses should proceed on a minimum mile length for any segment (for example, Rock Creek Road between 12th Avenue NE and 24th Avenue NE or Porter Avenue between Franklin Road and Tecumseh Road)
- Now, look at the number of collisions occurring on each of the segments. You will have to look at how the collision is coded. For example, it may be coded at the intersection of Franklin Road and 48th Avenue NE, but it may have occurred on 48th Avenue NE 1000-feet north of Franklin Road. In this example, the collision belongs to 48th Avenue NE and not to Franklin Road.
- Again, separate the segments into the city ward they fall into
- Rank the segments from largest number of collisions to the fewest for each Ward (again, there should be 8 separate lists)

Aggregating Volumes with Collisions

- Rank only those aggregated segments for which we have both volume information and collision information
- Combine the numbers for each segment (for example, if a segment ranks second in terms of the most collisions and the v/c ratio ranks fourth in terms of v/c ratio from high to low, the combination number for this segment would be 2+4 = 6)
- Rank the segments from high to low for each Ward (again, this will result in 8 separate lists)

- All rankings will be utilized by city staff to compile priorities for future Bond projects, etc.

Process Replication

In the future, the City of Norman will periodically repeat this analysis as new information becomes available. Traffic count data and collision data will be updated over time, and City ward boundaries or other supporting datasets may also change. An ideal submission will therefore include a process or tool that allows City staff to reproduce the analysis and generate updated priority segment rankings using new datasets with minimal manual effort.

An ideal submission will:

Be capable of using the raw data sets provided or include an automated method for preparing, organizing, and cleaning the data for analysis. City staff should not be expected to perform extensive manual data cleaning before using the tool. Teams should consider whether removing or simplifying data during the cleaning process could result in the loss of useful information contained within police collision reports, traffic count records, or other source data.

Be designed to accommodate future changes to the datasets. For example, columns may be added, removed, renamed, or shifted, and the number of rows will change as additional years of data are collected. The process should allow City staff to make relatively simple adjustments when the structure of the source data changes.

Minimize the amount of specialized software, programming knowledge, or manual processing required by City staff to reproduce the analysis.

Consider compatibility with software currently available to the city. The city already uses macro-enabled Excel workbooks to automate certain processes, so solutions that can operate within Excel or other readily available City software may be easier to maintain and reuse. Solutions that rely on specialized software, external services, or platforms not currently available to the city should consider how those requirements may affect the City's ability to use the tool in future years. Reach out to us if you need to ask a question about compatibility.

The goal is to provide the city with a repeatable process that allows staff to replace or update the source datasets and efficiently reproduce the roadway segment analysis, ward-level rankings, and final aggregated priority lists.

Recommendations

- We are looking for a ranked list of all segments for which we have data for each of the eight wards
- How you get to those lists may be more important to some judges than it might be to the City of Norman
- We are only looking for the ranked lists. We are not looking for you to make any recommendations or to spend time trying to figure out what may or may not need to be done. Once we have the list and select the Bond Projects, there will be engineering studies that will be conducted to determine the magnitude and potential cost of improvements to each segment.
- We are also looking for a system that we could reuse in future years to obtain similar results. So, your procedure needs to be able to be replicated.

Contact Us

With any questions you have during the competition, please reach out to us. You can reach any of us at:

- Awet.Frezgi@NormanOK.gov
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Questions:

- We need help defining candidate locations for future transportation bond programs. We know where proposed developments occur and that is how we have chosen candidate projects in the past. However, this approach ignore what may be an existing problem that need to be addressed. What are the top 20 locations in Norman when looking at the combination of collisions occurring along with traffic volumes?
- What are the top 20 locations in Norman based solely on traffic volumes?
- What are the top 20 locations in Norman based solely on collisions that have occurred?
- Are we able to have a list of the top 20 locations in Norman vs. volumes, vs. collisions, and vs. an aggregate of the two for each of the past five years so that we can identify locations that have been a problem each year rather than appearing on a top 20 list for only one year?

- Past bond programs were structured to recommend projects in each Ward to try to maximize the likelihood of a successful outcome in a voter election. Can the results be provided in a map format overlaid on the City Ward Map to make sure we are getting good representation for each Ward? This may require that we have to look at more than the top 20 locations in the City. In mapping these collisions/volumes/collision-volume aggregate, we would envision some sort of heat map for demonstration purposes.